HPS AVIATION

The Next Step to the Sky

Revolutionary propulsion technology inspired by nature,
designed for the future of aviation.

Why Propellers Are Holding Us Back

Noise Pollution

Traditional propellers generate excessive noise (85-100 dB), limiting urban applications and causing noise pollution.

Inefficient Energy Use

Current electric propulsion systems waste energy through turbulence and drag, limiting range and performance.

Limited Control

Conventional propulsion offers limited degrees of freedom, requiring complex mechanical systems for maneuverability.

Bulky and Exposed Design

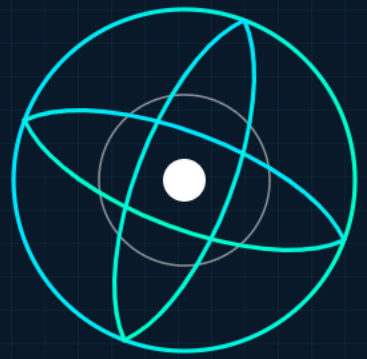
Propellers are large and usually exposed, which:
Increases drag
Poses safety hazards to people or nearby objects
Limits vehicle compactness and urban landing use

Humming Propulsion System (HPS)

Inspired by Nature

Our revolutionary propulsion system mimics the efficient wing movements of hummingbirds, combining rotational and oscillatory motion patterns.

- ✓ Full 6 degrees of freedom thrust control
- ✓ Ultra-quiet: < 65 dB
- ✓ 25% more efficient than propellers
- ✓ Enclosed design for enhanced safety



CROW: The First Hydrogen-Powered Flying Car

Technical Specifications

- Speed: 500 km/h
- Range: 1500 km
- Altitude: 12,000 ft
- Capacity: 5 passengers + cargo
- Can land on water and drive on roads
- Noise Level 65 dB at 5 meters
- 10 safety systems
- Hydrogen-Powered

CROW is the world's first flying car designed to match the Hollywood dream — a true car that flies, combining the look and feel of a futuristic vehicle with the power and agility of advanced aviation technology.

Powered by the revolutionary Humming Propulsion System (HPS), CROW is compact, efficient, and versatile, built for both urban and intercity mobility.

Market Vision

Air Taxis

Urban mobility revolution with on-demand aerial transportation, reducing commute times by 80%.

Cargo Drones

Autonomous delivery systems for last-mile logistics and emergency supplies to remote areas.

Flying Hotels

Luxury aerial accommodations offering unique travel experiences and unprecedented mobility.

Accessible Aviation

Our vision extends beyond technology to democratize flight. With intuitive controls and autonomous capabilities, HPS Aviation will make personal air travel accessible without extensive pilot training.

$980B
Global Market by 2035

43%
Annual Growth Rate

2028
Commercial Launch

Roadmap to Revolution

Phase 1: Concept & Research (2014 – 2020)

Fundamental research and technology development. Patent portfolio established with 12 core technologies.

Phase 2: Advanced Prototyping & Company Formation (2021 – First Half 2025)

Full-scale working prototype of the CROW flying car. Testing and certification process begins.

Phase 3: MVP Development & Lab Construction (Second Half 2025)

Maiden voyage of the CROW. Public demonstrations and regulatory approval process.

Phase 4: CROW – First Flying Car Prototype (2026 – 2027)

Production facility established. Supply chain secured and manufacturing begins.

Phase 5: Market Entry & Early Sales (2028 – 2029)

Commercial launch of the CROW flying car. Initial deliveries to strategic partners and early adopters.

Phase 6: Expansion & Diversification (2030 and Beyond)

Scale the business by expanding HPS integration across multiple flying platforms beyond CROW

2014 2020 2021 2025 2026 2027 2028 2029 2030

Phase 1: Concept & Research (2014 – 2020)

2014 – Initial idea inspired by bird and insect flight
2015 – Mathematical modelling of hummingbird wing patterns using MATLAB & ANSYS
2016 – Resigned from military R&D to dedicate full-time focus
2017–2020 – Developed and tested multiple mechanical prototypes
Built CAD, CFD, and 3D-printed models
Achieved 25% higher efficiency compared to traditional propulsion
Validated full 6-DOF control without tilting the propulsion system

2014 2015 2016 2017 2020

Phase 2: Advanced Prototyping & Company Formation (2021 – First Half 2025)

- Simplified HPS design for manufacturability
- Integrated hybrid control (rotational + oscillatory motion)
- Finalized scalable axial-flux electric motor architecture
- 2024 – Designed full MVP model
- Registered HPS Aviation and prepared for equity crowdfunding

2021 — 2025

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Phase 3: MVP Development & Lab Construction (Second Half 2025)

- Target funding: $1.5M
- Build engineering team (CFD, mechanical, business)
- Set up R&D lab + acquire equipment (5-axis CNC, lathe, metal 3D printer)
- Complete MVP for HPS propulsion system within 6 months
- Finish Patents registration + certification & safety testing

2025

Phase 4: CROW – First Flying Car Prototype (2026 – 2027)

- Begin full-scale build of CROW
- Materials: carbon fiber + bamboo composite
- Power: Hydrogen fuel cells + safety batteries
- Integrate 10 advanced safety systems (including Aerospike emergency landing)
- Flight tests, government compliance, and certification
- Showcase to investors, public, and strategic partners

2026 — 2027

Phase 5: Market Entry & Early Sales (2028 – 2029)

- Launch limited production of CROW
- Price: $500K – $1M based on configuration
- Build early adopter community (private owners, corporates, tourism)
- Begin licensing HPS propulsion to other platforms (eVTOLs, amphibians)
- Establish after-sales, service, and maintenance network

2028 — 2029

Phase 6: Expansion & Diversification (2030 and Beyond)

Develop new air mobility vehicles:
Cargo drones
Flying taxis
Modular flying hotels
Integrate HPS in:
Defense applications
Construction air platforms
Human-scale drones
Explore open-source HPS licensing model
Global expansion and strategic joint ventures

2030

Crowdfunding Round – Own a Share in the Future of Aviation

Campaign Overview:

- **Funding Goal**

  $1,500,000

- **Share Price**

  $1 per share

- **Pre-Money Valuation**

  $15,000,000

- **Equity Offered**

  10% of HPS Aviation

Use of Funds

- **Build the MVP of Humming Propulsion System (HPS)**
- **Strategic Partnerships**
- **Construct R&D and prototyping lab in Abu Dhabi**
- **Acquire key equipment (3D printer, CNC, lathe)**
- **Hire full core team: engineers, CFD, business dev**
- **Begin advanced development of CROW flying car**
- **Patent filing, certification, and early marketing**

Why Invest Now

- **Patent Portfolio**

  12 patents secured for our core technologies, creating a strong competitive moat.

- **Strategic Partnerships**

  Agreements with key aerospace suppliers and hydrogen fuel cell manufacturers.

- **Regulatory Advantage**

  Working closely with aviation authorities to establish certification pathways.

- **Market Timing**

  Perfect intersection of hydrogen technology maturity and growing demand for sustainable aviation.

Investor Perks (based on tiers):

- **Digital investor certificate & name in the credits**
- **Exclusive merch (CAP, HPS shirt)**
- **Access to flight sim & development updates**
- **Early reservation or discount for CROW when launched**
- **VIP invite to MVP launch event in Abu Dhabi**

Join Our Journey

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Founder

Ahmed Hamza

CEO & Chief Engineer

SCHEDULE MEETING

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